

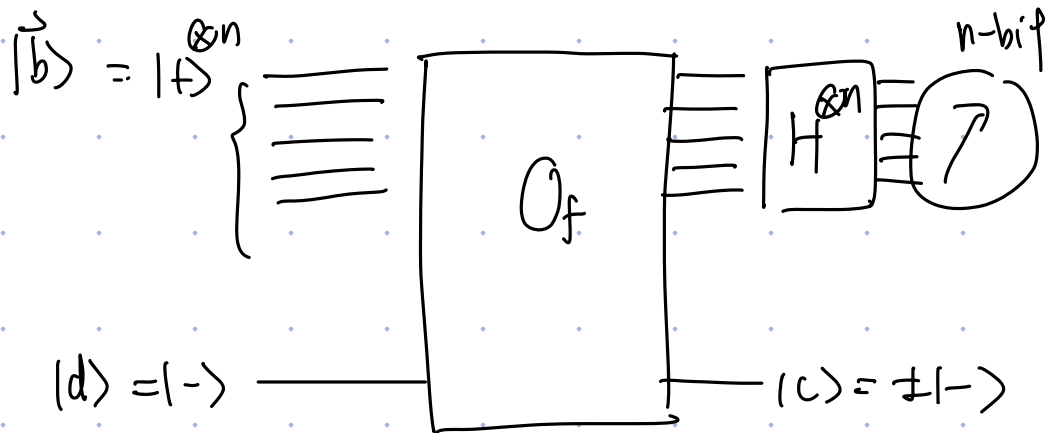
Lecture 19

Goals: ① Bernstein Vazirani: $f(\vec{x}) = \vec{x} \cdot \vec{u}$

② Simon's Problem.

(in BQP but not BPP)

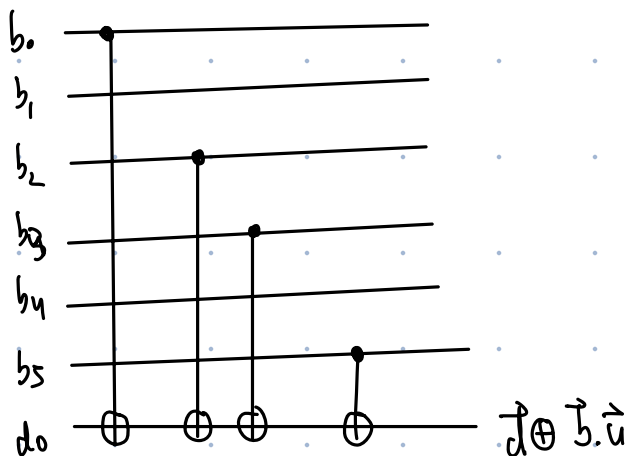
↳ exponential advantage even over probabilistic classical algs.

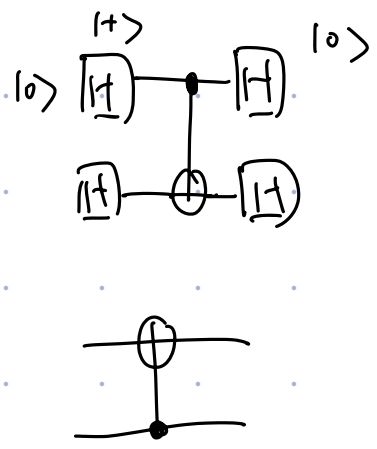
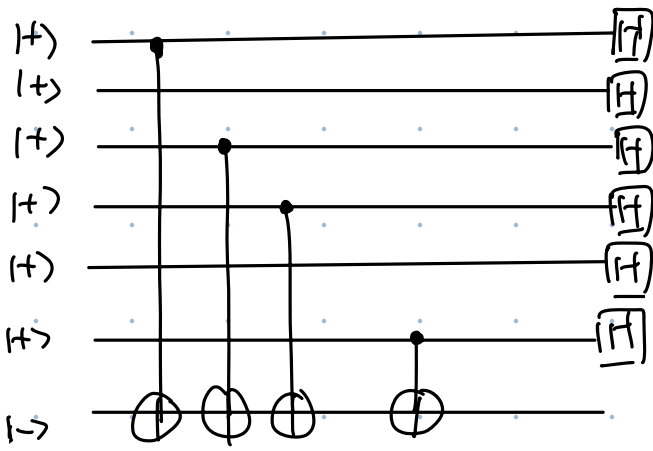


$$f: \{0,1\}^n \rightarrow \{0,1\}$$

$$f(\vec{x}) = \vec{x} \cdot \vec{u} \begin{cases} \text{if } \vec{u} = 0, f(\vec{x}) = 0 \\ \text{if } \vec{u} \neq 0, f(\vec{x}) \text{ is balanced.} \end{cases}$$

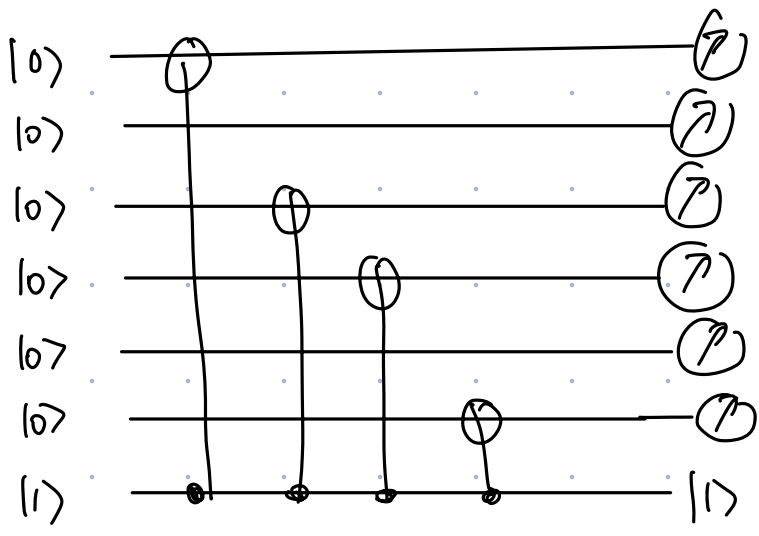
B-V Oracle for $n=6$ and $\vec{u} = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$





$H|+\rangle = |0\rangle, H|-\rangle = |1\rangle.$

$H^2 = I$



→ this true in general?
 is there a basis for which a circuit looks trivial?

Simon's Problem

↳ Related to the birthday Problem:

1. Given $\vec{f}: \{0,1\}^n \rightarrow \{0,1\}^n$ ← new: n-bits to n-bits
2. Guarantee that $\vec{f}$ is either one-to-one or two-to-one. (injective)
3. and guarantee $f(\vec{x}_1) = f(\vec{x}_2) \iff \vec{x}_2 = \vec{x}_1 \oplus \vec{b}$

Task: find $\vec{b}$

Ex: $n=2$ and $\vec{b}=(10)$

Table 3.2

$\vec{x}$	$\vec{x} + \vec{b}$	$f_1(x)$	$f_2(x)$	$f_3(x)$
00	10	00	10	01
01	11	11	01	11
10	00	00	10	01
11	01	11	01	11

pairs with
same
birthday

$(\Rightarrow)$ pairs of x_i
with same value.

Classical Query Complexity assume $\vec{b} \neq \vec{0}$

$\rightarrow$ if we could find 2 bitstrings $\vec{x}_1, \vec{x}_2$
s.t. $f(\vec{x})$, then

$$\vec{x}_1 \oplus \vec{x}_2 = \vec{x}_1 \oplus (\vec{x}_1 \oplus \vec{b}) = (\vec{x}_1 \oplus \vec{x}_1) \oplus \vec{b} = \vec{b}$$

worst case:

$\left\{ \begin{matrix} x_0 \\ x_1 \\ \vdots \end{matrix} \right\}$ one group

$\left\{ \begin{matrix} x_{2^{n/2}-1} \\ \vdots \\ x_{2^n-1} \end{matrix} \right\}$ the partner of that group.

Classical complexity:

$$\frac{2^n}{2} + 1 = 2^{n-1} + 1$$

→ Bob can choose the x 's randomly,

Failure probability after a number of tries.

↳ fail to find a match.

$$\frac{2^{n-1}}{2^n} \cdot \frac{2^{n-1}-1}{2^n-1} \cdot \frac{2^{n-1}-2}{2^n-2} \dots$$

$x_1, x_2, \dots$

$P_1 \leftarrow \text{finding a partner}$

$$P_1 = \frac{1}{2^n-1}$$

$x_1, x_2, \dots, x_3$

$$P_2 = \frac{1}{2^n-2} + \frac{1}{2^n-2} = \frac{2}{2^n-2}$$

$$P_j = \frac{j}{2^n-j}$$

$P_j = \frac{(j-1)}{2^n-(j-1)}$ convention in course notes.

j^{th} one out j ones already out.

$$P_n = \prod_{j=1}^n (1 - P_j) = \prod_{j=1}^n \left(1 - \frac{j}{2^n-j} \right)$$

$$= \prod_{j=1}^n \left(\frac{2^n-j}{2^n-j} \right) = \prod_{j=1}^n \left(\frac{1 - \frac{j}{2^n}}{1 - \frac{j}{2^n}} \right)$$

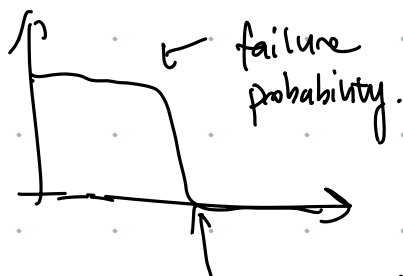
$$e^x \approx 1+x$$

$$= \prod \left(\frac{e^{-\frac{2j}{2^n}}}{e^{-\frac{j}{2^n}}} \right) = \prod \left(e^{-\frac{j}{2^n}} \right)$$

$\pi \times 10^7$
seconds
in a year

$$= e^{-\frac{1}{2^n} \sum_{j=1}^n j} = \underbrace{e^{-\frac{1}{2^n} \frac{n(n+1)}{2}}}_p$$

$$\sim \frac{1}{e} \text{ when } \frac{n(n+1)}{2} = 2^n$$



$$n(n+1) = 2^{n+1}$$

$$\log_2[n(n+1)] = n+1$$

$n \sim 2^{n/2} \rightarrow$ still exponentially big.